AUDIT COMPETITION SUMMARY REPORT

ALCHEMIX V3

JAN 2026

INDEX

OVERVIEW

Immunefi Audit Competitions are special, time-limited events that supercharge the reach and visibility of programs to our whitehat community.

From 14 October 2025 to 4 November 2025, the Alchemix V3 Competition offered up to $100,000 USD in rewards for security researchers.

Immunefi's Discord server hosted a channel for enhanced, two-way communication between whitehats and the Alchemix team, improving feedback and response times. Managed Triaging was also activated for the duration of this event, streamlining the resolution process for incoming bug reports.

During this event, 67 critical bugs, 178 high, 80 medium bugs, 155 low, and 47 insight reports were found on the target contracts. A total of 298 security researchers participated.

Alchemix distributed the $100,000 reward pool to 527 of the very best submissions for the security researchers' valiant efforts, including "Insight" submissions scored on a rating system that takes into account levels of:

1) Security best practices

2) Code optimizations and enhancements

3) Architectural decentralization and composability

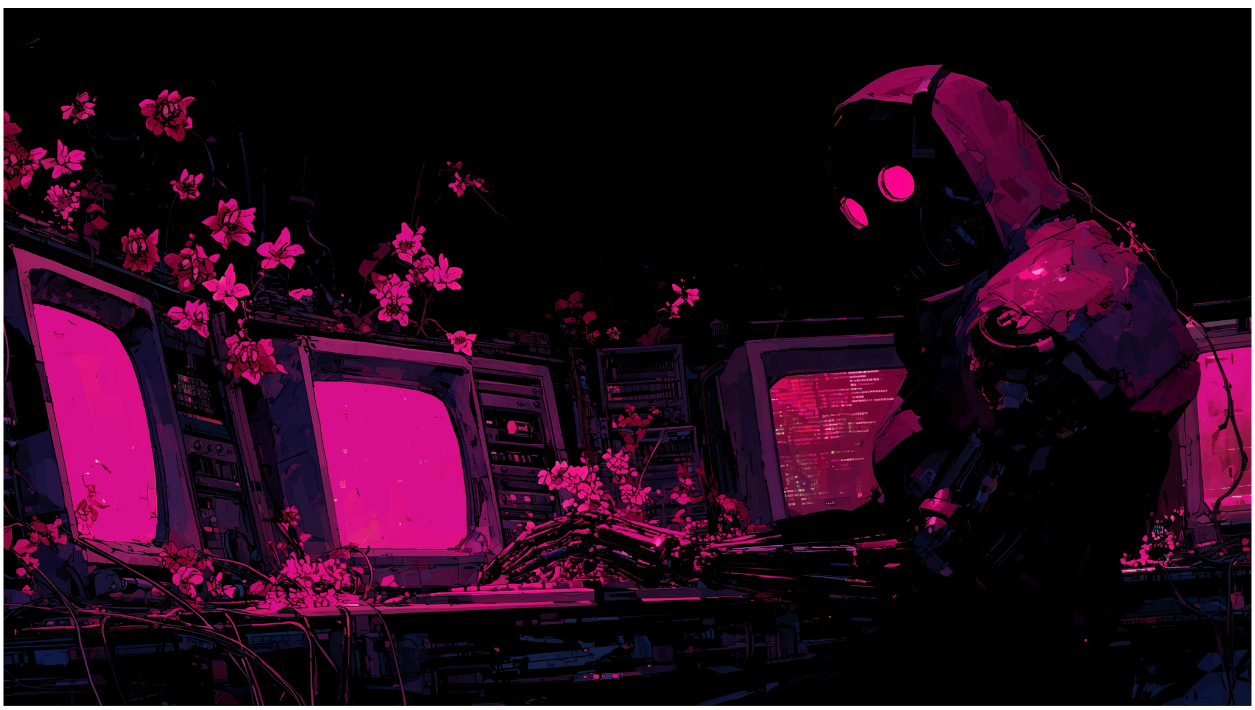
4) Documentation improvements

PROJECT NAME INTRO

Alchemix is your unified platform for saving, earning, borrowing, and fixed-term fixed-yield opportunities—all in one place. Built on years of iteration since launching the original self-repaying loan in 2021, Alchemix v3 brings all three pillars together with a smarter, more flexible design.

SCOPE OF ASSETS

The target assets in scope for the Audit Competition were Alchemix's smart contract assets. The total nSLOC was 2612.



SUMMARY

Duration: Three weeks	Comp Date: 14 Oct 2025 - 4 Nov 2025	Rewards Pool: $100,000	nSLOC: 2612
Submitted reports: 952	Security researchers: 298	Valid vulnerabilities: 480	Insight reports: 47

TOTAL WHITEHAT PARTICIPATION

TOTAL RESEARCHERS	298
PAID RESEARCHERS	191

LEADERBOARD

Position	Reward	Username	Valids	Insights
1	12446	farismaulana	15	1
2	8714	zeroK	5	0
3	7780	niroh	9	0
4	6997	magtentic	3	2
5	6748	Paludo0x	4	1
6	6698	shadowHunter	1	0
7	3632	gizzy	5	0
8	3557	pirex	2	0
9	3556	arturtoros	2	0
10	3556	MahdiKarimi	1	0
11	2554	silver_eth	3	0
12	1868	0xPhantom	7	0
13	1832	kenzo	4	0
14	1756	XDZIBECX	5	0
15	1556	fullstop	5	1
16	1537	Tadev	8	0
17	1537	Icon0x	1	1

18	1533	JoeMama	6	1
19	1443	algizsec	2	0
20	1119	pashap9990	7	2
21	1096	ox9527	7	0
22	1057	hashbug	3	0
23	909	godwinudo	6	0
24	902	T0nraq	3	0
25	899	Brainiac5	3	0
26	889	al0x23	2	0
27	887	dobrevaleri	8	2
28	864	blacksaviour	1	0
29	729	terrah	3	0
30	636	PotEater	6	5
31	611	Outliers	5	0
32	567	aua_oo7	2	0
33	437	aman	6	0
34	437	teoslaf1	8	0
35	435	MentemDeus28	3	0
36	420	blackdruiid	1	0
37	403	enoch	2	1
38	384	0xdeadmanwalking	10	1
39	360	Razkky	2	0
40	323	oxrex	6	1
41	305	nem0thefinder	9	1
42	291	Cryptor	3	0

43	255	manvi	2	1
44	211	algiz	2	1
45	195	SOPROBRO	5	0
46	194	OxPrince	0	2
47	175	joicygiore	7	1
48	171	damdam0249	3	0
49	158	a16	5	0
50	156	pindarev	2	0
51	147	luc1jan	2	0
52	147	flora	1	0
53	137	Smartkelvin	4	0
54	135	spongebob	4	1
55	134	Cyborg	5	0
56	130	mohitisimmortal	2	0
57	117	sol_4th05	1	1
58	116	Snuggle	1	2
59	116	wylis	1	2
60	116	chief_hunter888	2	2
61	109	Anirruth	4	1
62	100	Ambitious_DyDx	2	0
63	99	X0sauce	2	0
64	99	j3x	1	0
65	99	Arkindyo	1	0
66	97	Davuka	1	1

67	96	Bluedragon	2	0
68	93	hunter0xweb3	4	1
69	92	InAllHonesty	2	0
70	87	Josh4324	6	1
71	78	Petrus	5	0
72	77	haidoka017	0	2
73	77	silencedogood	1	0
74	77	sus_bandicoot	1	0
75	58	SAAJ	1	1
76	58	Purpledragon	0	1
77	58	rbd3	0	1
78	58	randomnpc	0	1
79	57	yesofcourse	6	0
80	57	legion	4	0
81	54	Orhuk1	3	0
82	54	pxng0lin	5	0
83	47	Codexstar	2	0
84	46	Jugger63	2	0
85	41	winnerz	2	0
86	40	0xvictorsr	1	0
87	39	jayx	3	0
88	39	kodyvim	0	1
89	39	tygra	0	1
90	39	lirezarazavi	0	1
91	38	Django	2	0
92	38	Impala53732	1	0

93	38	InquisitorScythe	1	0
94	38	Tarnishedx0	3	0
95	37	IronsideSec	2	0
96	37	kaysoft	4	0
97	36	dldLambda	4	0
98	32	ByteKnight	3	0
99	31	ayden	1	0
100	30	humaira45	5	0
101	29	dray	4	0
102	27	Ratt13snak3	5	0
103	26	dizaye	2	0
104	25	fawarano	1	0
105	21	Idealz	2	0
106	20	griffin	1	0
107	20	unique	1	0
108	17	emmac002	2	0
109	17	x0xmechanic	2	0
110	16	failsafe_intern	3	0
111	15	oct0pwn	3	0
112	15	securehash1	2	0
113	15	nvalkov	1	0
114	13	riptide	3	0
115	13	Pro_King	2	0
116	13	grearlake	1	0
117	13	theboiledcorn	1	0

118	13	vivekd	2	0
119	12	zcai	4	0
120	12	xanony	2	0
121	11	Bizarro	3	0
122	11	mzfr	2	0
123	11	Another	2	0
124	10	Pig46940	1	0
125	9	Lion47624	2	0
126	9	Coachmike	1	0
127	9	omarAli001	1	0
128	9	resosiloris	1	0
129	9	bigbear1229	1	0
130	7	Bug82427	2	0
131	6	LogicalJosselyn798	1	0
132	6	ZenHunter	1	0
133	6	Diavol0	4	0
134	5	DeoGratias	2	0
135	3	niffylord	3	0
136	3	auditagent	2	0
137	3	KiLl3rX	1	0
138	3	ayeslick	1	0
139	3	edantes	2	0
140	3	ihtishamsudo	4	0
141	3	HandsomeEarthworm6	3	0

142	3	Pataroff	2	0
143	3	Ibukun	1	0
144	3	Bear36435	1	0
145	2	ibrahimatix0x01	3	0
146	2	Tomioka	2	0
147	2	TyroneX	2	0
148	2	Aiden	1	0
149	2	Tee0x	1	0
150	2	iAfrika	1	0
151	1	call_me_rp	1	0
152	1	Novathemachine	2	0
153	0	rshackin	3	0
154	0	unineko	3	0
155	0	Kissiahmyo	2	0
156	0	DashBug	1	0
157	0	JavaScript36142	1	0
158	0	IShiftOnBlue	1	0
159	0	Johnyfwesh	1	0
160	0	konvati	1	0
161	0	EagleEye	2	0
162	0	ENIGMA	1	0
163	0	akioniace	1	0
164	0	liae	1	0
165	0	Eagle_Eye	1	0

166	0	KKam86	1	0
167	0	pikachu0203	1	0
168	0	silverologist	1	0
169	0	Y4nhu1	1	0
170	0	Ekene	1	0
171	0	acnologiac	1	0
172	0	Bx4	1	0
173	0	Vanshika	1	0
174	0	rand	1	0
175	0	Khay3	1	0
176	0	Shivansh	1	0
177	0	madman	1	0
178	0	s_a_l_e_m	1	0
179	0	cmds	1	0
180	0	vah_13	1	0
181	0	gor97	1	0
182	0	DeusVult	1	0
183	0	TianYu4n	1	0
184	0	[redacted]	0	1
185	0	[redacted]	1	0
186	0	[redacted]	5	0
187	0	[redacted]	1	0
188	0	[redacted]	2	0
189	0	[redacted]	1	0
190	0	[redacted]	1	0
191	0	[redacted]	1	0

TOP 5 REPORTS

claimRedemption would not return all alAsset that is not get converted to MYT in some case

Report number: 56702 Submitted by: @farismaulana

Target:
https://github.com/alchemix-finance/v3-poc/blob/immunefi_audit/src/Transmuter.sol

Impacts: Permanent freezing of funds

Program Action: Confirmed as critical severity.

Report Excerpt:

`claimRedemption` function in the `Transmuter` is designed to handle cases where the system can't obtain enough yield tokens (MYT) to fulfill a user's entire claim. It correctly caps the amount of MYT sent to the user. but the function fails to refund the corresponding portion of the user's synthetic asset (`alAsset`) that would be burned but not converted. resulting in a permanent loss of funds for the user, as their `alAsset` is destroyed without them receiving the equivalent value in yield tokens.

Incorrect boundary condition in queryGraph leads to systematic under-earmarking and transmuter redemption fund loss

Report number: 56732 Submitted by: @enoch

Target:
https://github.com/alchemix-finance/v3-poc/blob/immunefi_audit/src/Transmuter.sol

Impacts: Permanent freezing of funds

Program Action: Confirmed as critical severity.

Report Excerpt:

The Alchemix V3 protocol implements a transmuter system where users can deposit synthetic tokens (alAssets) to eventually receive yield-bearing tokens. The transmuter tracks position maturation through a staking graph data structure that records active stakes across block ranges.

The `AlchemistV3` contract maintains an earmarking mechanism to allocate borrower debt for redemption by transmuter users. The earmarking process queries the transmuter's staking graph to determine how much active stake should be converted into earmarked debt, which directly determines the redemption capacity available to transmuter users.

Liquidation Fee Overdraft Drains Pooled Collateral

Report number: 56365 Submitted by: @failsafe_intern

Target:
https://github.com/alchemix-finance/v3-poc/blob/immunefi_audit/src/AlchemistV3.sol

Impacts: Direct theft of any user funds, whether at-rest or in-motion, other than unclaimed yield

Program Action: Confirmed as critical severity.

Report Excerpt:

The `_liquidate` function pays liquidators a repayment fee that exceeds what was deducted from the liquidated account, stealing the shortfall from the protocol's pooled MYT collateral shared by all users.

When an account with earmarked debt is liquidated and the debt fully clears, `_resolveRepaymentFee` calculates the fee but only deducts `min(fee, collateralBalance)` from the account:

Repay removes earmark, meant to be reducing debt while collateral is still reduced

Report number: 57916 Submitted by: @JoeMama

Target:
https://github.com/alchemix-finance/v3-poc/blob/immunefi_audit/src/AlchemistV3.sol

Impacts: Direct theft of any user funds, whether at-rest or in-motion, other than unclaimed yield

Program Action: Confirmed as critical severity.

Report Excerpt:

When a user repays a portion of his debt, the `repay` function will remove the earmark, but keep the collateral position, when `sync` is called afterwards, it will **wrongly** remove some of the users collateral but not remove the debt because earmark is 0.

- One issue is that `Repay` is paying off the debt that would normally been removed by sync. (up to the earmark amount)

- The second issue is that it makes the user lose a portion of its collateral while not reducing debt during the next sync.

function repay(uint256 amount, uint256 recipientTokenId) public returns (uint256) {

.... snip ...

account.earmarked -= earmarkToRemove;

}

Integer Division Precision Loss in normalizeDebtTokensToUnderlying Leads to Permanent Collateral Locking

Report number: 57053 Submitted by: @enoch

Target:
https://github.com/alchemix-finance/v3-poc/blob/immunefi_audit/src/AlchemistV3.sol

Impacts: Permanent freezing of funds

Program Action: Confirmed as critical severity.

Report Excerpt:

A critical precision loss vulnerability exists in the AlchemistV3 contract's debt-to-underlying conversion logic. When a user or attacker repays a very small amount of debt (an amount less than the underlyingConversionFactor), the integer division in normalizeDebtTokensToUnderlying rounds the amount of collateral to be "freed" down to zero. This causes a critical state desynchronization: the user's debt is correctly reduced, but their internal rawLocked collateral accounting is not. This desynchronized state acts as a "trap," which is later triggered by a global redeem event. When triggered, the contract's _sync logic incorrectly seizes a portion of the user's actual collateralBalance to cover global redemptions, leading to a permanent and irrecoverable loss of user funds.

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